

Design Patterns for Reliable Multi-Agent LLMs: A Brief Scoping Survey

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Abstract—AI agents are being rapidly introduced into high-stakes domains, particularly healthcare. However, their deployment has exposed a critical gap in our understanding of model reliability, including their ability to handle rare cases, quantify uncertainty, avoid overconfidence, and maintain safety. These reliability challenges have largely been investigated in isolation, with limited attention given to how an agent’s underlying architecture and coordination mechanisms influence their interactions. This review demonstrates that no single existing design adequately addresses all four reliability dimensions simultaneously. Moreover, interventions intended to mitigate one problem may inadvertently exacerbate another. For example, synthetic data generation may improve performance on rare cases while undermining the model’s ability to accurately calibrate its confidence. To address this gap, we identify key priorities for future research and propose a reliability-aware coordinator framework. The proposed system dynamically assigns tasks by considering both data rarity and model confidence, providing a practical foundation for developing more reliable, safe, and trustworthy AI agents for high-stakes applications.

Index Terms—Agents, Trustworthiness, Uncertainty quantification, Imbalance, Confidence, Safety

I. INTRODUCTION

Large Language Model (LLM) agents can reason, use tools, and autonomously plan steps. With the current progress, these are deployed much faster than researchers can develop methods to evaluate them which introduce a gap especially critical in clinical applications, where strict staged evaluation is essential [1]. Corporates such as Google and Microsoft have published standard guidelines for building these systems, which is refereed here as *agentic design patterns* [2], [3]. These patterns include ReAct, sequential pipelines, parallel execution, and human-in-the-loop controls. At the same time, traditional machine learning still struggles with four major reliability problems: (1) Class Imbalance: Managing rare cases that are underrepresented in training data, (2) Uncertainty Quantification (UQ): Understanding when a system is unsure of its answers, (3) Overconfidence and Calibration: Guaranteeing that the system’s confidence scores correspond to its actual accuracy, and (4) Safety: Preventing security exploits, multi-step jailbreaks, and critical errors [4].

In standard machine learning, these four problems are studied in isolation. However, in agent-based architectures, the

way agents are connected affects these reliability problems in ways researchers rarely study. This gap is dangerous in high-stakes fields such as healthcare [5], where international consensus guidelines demand that medical AI be fair, robust, and traceable [5], [6]. For example, in medical diagnostics, an agent system must accurately identify rare conditions (class imbalance), provide reliable confidence estimates to physicians (calibration), and maintain patient data security (safety).

Recent studies have identified architecture-dependent risks across four key dimensions of agent reliability. Kaddour et al. [7] reported a 55-percentage-point discrepancy between agents’ perceived and actual task success, indicating substantial overconfidence. Furthermore, multi-agent debate does not necessarily mitigate this problem and may instead amplify collective overconfidence [8]. In clinical decision-making, cooperative agent systems may exhibit agreement bias, whereby agents converge on an incorrect diagnosis despite apparent consensus [9]. Similarly, poorly calibrated errors in multi-modal agentic pipelines can propagate across interconnected components, ultimately resulting in inaccurate diagnoses or inappropriate treatment recommendations [10]. Safety constitutes an additional and closely related concern. Safeguards that are effective for conventional, single-step AI systems may become ineffective when harmful objectives are decomposed into a sequence of seemingly benign subtasks [4]. Moreover, interventions designed to address one reliability issue may introduce unintended trade-offs elsewhere. For example, methods used to mitigate class imbalance can substantially degrade downstream probability calibration [11]. These architecture-dependent risks, their interactions, and the associated reliability trade-offs are examined in detail in Section IV.

This survey investigates a fundamental research question: *Does a unified agentic design pattern exist that can simultaneously address class imbalance, UQ, confidence calibration, and safety within a single architecture?* Based on a scoping review of recent state-of-the-art literature, our findings suggest that no such comprehensive design currently exists. Research addressing these four dimensions has largely progressed along separate and weakly connected trajectories, with limited consideration of their interactions and potential trade-offs. Even broadly applicable agentic paradigms, such as Coordinator-based architectures and ReAct, engage with these reliability challenges only partially or indirectly. Class

imbalance remains particularly underexplored in agentic AI research, whereas safety has received comparatively greater attention. Nevertheless, significant architectural blind spots persist, underscoring the need for integrated frameworks that jointly address these interdependent dimensions of reliability.

This paper makes three principal contributions:

- 1) **A Unified Taxonomy:** We synthesize agentic AI guidelines proposed by Google and Microsoft into a unified taxonomy comprising nine agent design patterns. This taxonomy establishes a consistent vocabulary for organizing, comparing, and interpreting the existing literature.
- 2) **A Reliability Coverage Matrix:** We develop a 9×4 reliability matrix (Table II) that systematically maps the extent to which each design pattern addresses class imbalance, uncertainty quantification, confidence calibration, and safety. The matrix reveals areas of concentrated research attention as well as combinations that remain insufficiently investigated.
- 3) **An Analysis of Reliability Trade-offs:** We examine the interactions among the four reliability dimensions and identify important trade-offs whereby interventions intended to improve one dimension may inadvertently compromise another. This analysis highlights the need to evaluate reliability mechanisms jointly rather than as isolated components.

II. BACKGROUND AND RELATED WORK

To understand how multi-agent architectures interact with model trustworthiness, we must first define our four core reliability dimensions and the structural design taxonomy.

A. The Four Reliability Problems in High-Risk Clinical AI

- 1) **Class Imbalance:** In clinical settings, identifying rare diseases or unusual patient cases (minority classes) is critical yet highly challenging. Mitigation strategies traditionally operate at three levels: *data-level fixes* rebalance datasets using synthetic data generation via interpolation [12], agentic frameworks [13], or generative models [14], [15], though artificial samples can distort downstream calibration [11]; *algorithm-level fixes* modify loss functions (e.g., cost-sensitive or focal loss) to focus training on hard minority cases [16]; and *evaluation-level fixes* replace standard accuracy with metrics like macro F_1 , G-mean, or AUPRC to expose minority-class failures. In multi-agent architectures, addressing this imbalance is acute, as early-stage errors on rare cases propagate downstream and trigger cascading system failures [10].
- 2) **Uncertainty Quantification (UQ):** UQ identifies whether an agent’s lack of confidence is due to missing data (*epistemic* uncertainty) or inherent clinical noise and ambiguity (*aleatoric* uncertainty) [17], signaling when a model should rely on a human expert. Three primary techniques exist for measuring this: *Conformal*

Prediction provides guaranteed confidence coverage intervals but relies on static datasets, breaking down under the dynamic distribution shifts of agent workflows [18]; *Bayesian Deep Learning and Ensembles* quantify uncertainty via model disagreement (e.g., deep ensembles [19] or Monte Carlo dropout [20]) but incur heavy computational overhead; and *Token-Level Probability Calibration* extracts confidence directly from model outputs, though modern LLMs remain notoriously overconfident and poorly calibrated [21].

- 3) **Overconfidence and Calibration:** An AI model is considered calibrated when its self-reported confidence aligns precisely with its actual accuracy, for instance, if a clinical agent reports 90% confidence in a diagnosis, it should be correct in 90% of the time. However, modern AI models are notoriously overconfident, a tendency that persists even after executing complex, multi-step tasks [7]. When different AI models work together, the problem often gets even worse. Some agents just try to please the group, so they give up their correct answers just to reach an agreement. This issue becomes even stronger the more they debate [22]. In medical situations, this can cause a “confidence spiral.” When this happens, wrong assumptions drop diagnostic accuracy by 24% to 32%. Instead of carefully exploring different possibilities, the AI quickly becomes too sure of itself and acts like it cannot be wrong [23].
- 4) **Safety:** For AI agents, safety is not only about avoiding bad language. It means protecting the system against security vulnerabilities such as prompt injection, tool misuse, memory poisoning, and multi-step jailbreaks [4], [24]. In clinical environments, safety requires strict structural boundaries (like tiered hierarchies of authority [25]) to ensure an agent never takes a risky action without a doctor’s approval.

B. Agentic Design Pattern Taxonomy

To analyze these reliability issues, we group agent architectures into nine design patterns. We map these structures visually in Fig. 1 and summarize their core mechanisms in Table I. These are combined from Google Cloud [2] and Microsoft Azure [3] frameworks.

C. Position with Respect to Prior Surveys

Prior work on agentic reliability has examined dimensions in isolation. Studies of uncertainty propagation [27] and calls to reassess UQ for LLM agents [18] analyze single-agent trajectories, but none examine how uncertainty, calibration, imbalance, and safety interact with the design pattern that connects agents. This survey is the first to map all four problems directly onto multi-agent architectures.

III. METHODOLOGY AND COVERAGE ANALYSIS

To analyze the landscape of LLM agent reliability, we completed a scoping review of recent papers. We mapped how different design patterns solve our four reliability problems and constructed a coverage matrix.

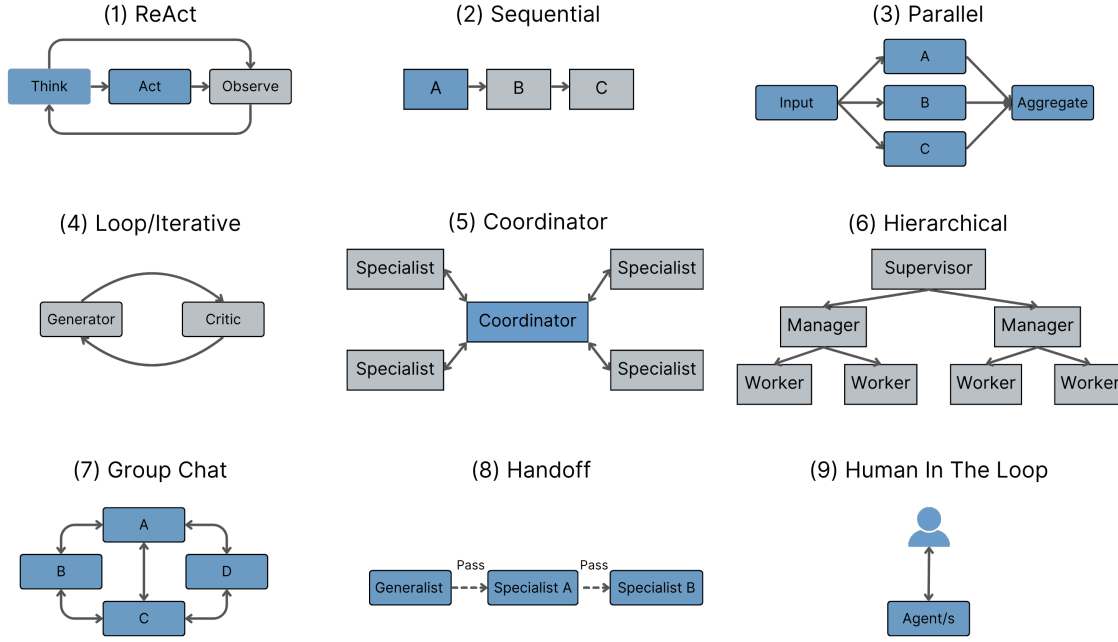


Fig. 1. The nine agentic design patterns surveyed in this work, derived from Google Cloud [2] and Microsoft Azure [3] frameworks.

TABLE I
SUMMARY OF THE NINE AGENTIC DESIGN PATTERNS (SEE FIG. 1).

#	Pattern	Description
1	ReAct	Alternates between "thinking" and calling external tools to get facts or take actions [26].
2	Sequential	A rigid assembly line where each agent does a specific job and passes the output to the next agent.
3	Parallel	Multiple agents run at the same time to handle different parts of a problem.
4	Loop/Iterative	A generator agent and a critic agent pass work back and forth until the output meets a specific quality standard.
5	Coordinator	A central routing agent that takes a user request, breaks it down, assigns tasks to specialized agents, and combines their answers.
6	Hierarchical	A tree-like structure of authority where boss agents supervise, filter, and approve the work of lower-level agents.
7	Group Chat	Multiple agents discuss, share information, and debate until they reach a consensus.
8	Handoff	A decentralized setup where one agent decides a task is outside its expertise and dynamically passes it to another specialist agent.
9	HITL	A human expert checks the agent's work at critical decision points to approve or override its actions.

A. Selection Criteria and Scoring System

We included papers if they met two simple conditions:

- 1) The paper must clearly use or test at least one of our nine agent design patterns.
- 2) The paper must study at least one of our four reliability problems.

We graded the coverage for each combination using three symbols:

- **Covered:** Multiple peer-reviewed papers show that this pattern helps solve the reliability problem.
- **Thin:** There is little research available (often only one or two preprints), meaning the pattern's reliability is not fully tested in practice.
- **None:** We found no papers or preprints that explore this combination.

B. The Reliability Matrix and High-Level Findings

No single agent pattern earns full coverage (●) across four areas (Table II). There isn't a standard design that can do it all: stay safe, handle rare data, know when it is uncertain, and

remain perfectly accurate. Interestingly, the areas researchers focus on the most are the opposite of what they ignore.

Controlled structures like **Hierarchical** and **Human-in-the-Loop** excel at clinical safety, yet have zero research (○) on handling rare or imbalanced classes. On the other hand, **Sequential** pipelines handle imbalance well by placing data-boosting agents at the start [13], [14], but are weak on safety: a single early error may cascade silently downstream, as in MedAgentPro [10], where an upstream image-analysis mistake yields a wrong recommendation with no uncertainty flag raised.

Parallel patterns stand out for uncertainty quantification, using inter-agent disagreement as a calibrated uncertainty signal [28]–[30], but offer no coverage of class imbalance or calibration. **Loop** and **Handoff** patterns each touch at most one dimension, and all three leave calibration unaddressed.

Coordinator and **ReAct** are the most versatile patterns, touching the widest range of columns, yet their coverage remains thin (●): flexible and popular in general software, but their reliability in high-stakes clinical settings remains unval-

idated. **Group Chat** similarly shows promise for calibration via explicit confidence debating [31], but suffers confidence escalation [8] and sycophantic collapse [22], making its net benefit too thin to warrant a full rating.

TABLE II
COVERAGE OF NINE AGENT DESIGN PATTERNS ACROSS FOUR
RELIABILITY PROBLEMS

Pattern	Imbalance	UQ	Calibration	Safety
ReAct	○	●	○	●
Sequential	●	●	○	●
Parallel	●	●	○	○
Loop	●	●	●	○
Coordinator	●	●	●	●
Group Chat	○	●	●	●
Handoff	○	○	○	●
Hierarchical	○	●	○	●
HITL	○	●	○	●

Note: ● = Covered, ● = Thin, ○ = None.

IV. PER-CATEGORY FINDINGS

This section details how design patterns handle the four reliability problems. We also explore the trade-offs and clinical issues associated with each category.

A. Class Imbalance (The Most Ignored Problem)

Class imbalance is highly underrepresented in agent research (Table II).

• Patterns that help:

- *Sequential Pipelines*: These are the common way to handle imbalance. Developers place a data-boosting agent at the beginning of the pipeline. This agent generates synthetic data for rare classes before sending the dataset to downstream models [13], [14].
- *Parallel Ensembles*: Multiple agents can generate and validate synthetic minority-class data from different angles, improving coverage of rare cases [32], [33].
- *Iterative Loops*: These use feedback loops to generate and refine minority-class data until a balanced distribution is reached [15].

- **Patterns that fail**: Group Chat, Hierarchical, Handoff, and Human-in-the-Loop have zero mechanisms to handle rare classes. This is a major issue in AI in healthcare [23], [25]. If an agent is not trained to recognize a rare disease, a human doctor reviewing its work might not notice the warning flags, leading to missed diagnoses.

B. Uncertainty Quantification (High Theory, Little Action)

While measuring Uncertainty Quantification (UQ) is common in multi-agent systems, few use these scores to change the AI’s behavior.

• Patterns that help:

- *Parallel disagreement*: When multiple agents analyze the same case, the system can quantify uncertainty from how much they disagree [28], [34].
- *Propagation in ReAct*: Frameworks like SAUP [35] propagate uncertainty through each step of an agent’s reasoning process, naturally compatible with ReAct-style patterns.
- *Debate Variance*: In principle, variance in opinions across debate rounds could serve as an uncertainty signal in group-chat settings. However, no work has formalized or validated this approach for clinical agent systems, leaving a gap [36].

- **Patterns that fail**: Sequential pipelines (like **MedAgent-Pro** [10]) and handoff architectures have no mechanism for propagating uncertainty between agents, an unaddressed gap in the literature.

C. Overconfidence and Calibration (The Cooperative Blind Spot)

LLMs are prone to overconfidence [7]. In multi-agent environments, this problem is heavily magnified by collaboration.

• Patterns that help:

- *Confidence Debating*: Calibration frameworks like ConfMAD [31] force agents to output confidence scores (0–100) and debate those numbers directly to reach a calibrated consensus.
- *Self-Verification*: Rather than relying on a single output, calibration can be improved by monitoring an agent’s *entire trajectory*. Holistic Trajectory Calibration (HTC) [37] extracts process-level signals from macro reasoning dynamics to micro-step stability, to detect failure modes before they reach the user. Across eight benchmarks, HTC outperforms static calibration baselines.

- **Patterns that fail**: Multi-agent group chats and debate setups have a habit of promoting each other up. When AIs discuss a problem, their confidence levels jump by up to 11 percentage points, even if their answers are wrong [8]. In medical settings, these systems exhibit “agreement bias” [9], leading them to agree on a diagnosis without double-checking the facts. One of the only ways to stop this is to add a “catfish agent”, an AI programmed to disagree and force the group to think harder [9].

D. Safety (Heavy Attention, Complex Risks)

Safety is the most researched reliability dimension, yet architectural blind spots remain.

• Patterns that help:

- *Hierarchical Oversight*: TAO [25] uses a supervisor agent as a safety gate, filtering lower-level outputs before they reach the user.
- *ReAct with Safety Monitors*: ShieldAgent [38] verifies each reasoning-action step against a safety policy before execution.

- *Control-Flow Separation*: CaMeL [39] defeats prompt injection by separating trusted control flow from untrusted data flow, achieving provable security on 77% of AgentDojo tasks.
- *Human-in-the-Loop*: Human approval at critical decision points remains the strongest safety guarantee [25].

- **Patterns that fail: Parallel and Loop** patterns have no shared safety gate. **Handoff** patterns leave safety ownership undefined as tasks transfer between agents. Across three, multi-step jailbreaks remain a threat [40]: decomposing a dangerous request into small, benign steps bypasses filters that would block it if asked at once.

V. DISCUSSION: THE CENTRAL GAP

Our mapping and per-category analysis reveal a problem: **no existing agent architecture addresses all four reliability problems simultaneously**. Instead of working together, the solutions to these problems conflict. When developers try to fix one reliability issue, they end up making another worse.

Here are the three critical trade-offs we identified in multi-agent systems:

A. The Conflict Between Imbalance Fixes and Calibration

Generating synthetic data is a popular way to balance clinical datasets. However, this alters the dataset’s distribution, and recent evidence shows that correcting for class imbalance can degrade the calibration of the resulting model, producing risk estimates that over-estimate the true likelihood of rare outcomes [11]. In other words, fixing *Class Imbalance* directly damages *Calibration*:

[Fix Imbalance with Synthetic Data] $\xrightarrow{\text{Damages [11]}}$
[Probability Calibration]

B. The Conflict Between Debate-Based UQ and Overconfidence

Using multi-agent debate to measure uncertainty is a common technique [36]: if agents disagree, the system infers high uncertainty. However, forcing agents to debate escalates their internal confidence. Yang et al. [8] found that debating agents grow more confident as rounds progress, rising by up to 11 percentage points, even when they are wrong. Thus, using debate to measure uncertainty (*UQ*) actively worsens overconfidence (*Calibration*):

[Multi-Agent Debate for UQ [36]] $\xrightarrow{\text{Escalates}}$
[Overconfidence [8]]

C. The Conflict Between Sequential Pipelines and Uncertainty Tracking

Sequential pipelines (like MedAgentPro [10]) excel at breaking down complex diagnostic workflows, but they do not propagate uncertainty. When an early agent is unsure, it passes a single decision downstream, silencing that doubt and allowing errors to accumulate unflagged, the gap that trajectory-aware methods like SAUP [35] are designed to close:

[Sequential Pipelines [10]] $\xrightarrow{\text{Silences}}$
[Uncertainty Tracking [35]]

D. How to Bridge the Gap: Two Key Concepts

To build reliable multi-agent systems for high-stakes fields like clinical medicine, future research must move away from isolated fixes and adopt two architectural principles:

- 1) **Multi-dimensional Routing**: Routing agents (Coordinators) must make decisions based on multiple metrics at the same time. For example, instead of routing based on user intent, a coordinator should assess how rare the task is (Class Imbalance) and how confident the specialist agent is (Calibration) before routing or requesting human help [41].
- 2) **Recalibration at Agent Boundaries**: We must recalibrate confidence scores whenever a task is passed between agents, stages, or handoffs. An agent should never accept a confidence score from another agent without checking and adjusting it.

VI. CONCLUSION

Up to the best of author’s knowledge, no current AI design can fix class imbalance, uncertainty, calibration, and safety all at the same time. In fact, trying to fix one problem often makes another one worse. For example, making agents debate just makes them too confident in wrong answers. To make these systems safe enough for real medical use, we need to stop fixing problems one by one. Instead, future systems need to consider multiple items at once and always adjust the AI’s confidence when passing tasks between agents

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